

Crystallization of the amorphous phase and martensitic transformations in multicomponent (Ti,Hf,Zr)(Ni,Cu)-based alloys

A. Pasko^{a,*}, V. Kolomytsev^a, P. Vermaut^b, F. Prima^b, R. Portier^b,
P. Ochin^c, A. Sezonenko^a

^a *Institute of Metal Physics NASU, 36 Vernadsky Blvd., 03680 Kiev, Ukraine*

^b *ENSCP, 11 rue Pierre et Marie Curie, 75231 Paris cedex 05, France*

^c *CECM CNRS, 15 rue Georges Urbain, 94407 Vitry-sur-Seine, France*

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Abstract

Shape memory materials with specific microstructure can be obtained from amorphous precursors. Rapidly solidified amorphous and crystalline–amorphous ribbons have been produced by planar flow casting for a number of multicomponent TiNi-based alloys of pseudobinary 50:50 and 55:45 compositions with substitutions of Zr, Hf, Nb for Ti and Cu, Co, Pd, Ag, Al for Ni. The glass transition and crystallization behavior of the amorphous phase as well as the martensitic transformations in the crystallized materials have been studied by differential scanning calorimetry. To control the crystal structure at different stages of crystallization, X-ray diffractometry and transmission electron microscopy were used. The effects of crystallization conditions and the resulting microstructural state on the martensitic transformation characteristics have been investigated. The transformation temperature intervals are shown to depend on the mean size of the parent phase crystals.

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1. Introduction

Structural phase transformations, which play an important role in changing the physical properties of solids and creating functional materials, can occur simultaneously or sequentially giving new opportunities for their research and application. TiNi-based shape memory alloys are widely used materials with martensitic transformations. Alloying the TiNi intermetallic compound with Cu improves its functional properties [1–3], while a sufficient addition of Hf or Zr increases the martensitic transformation characteristic temperatures [4–6]. On the other hand, under certain conditions some metallic materials (as alloys of Zr, Hf and Cu) can be easily obtained in an amorphous

state [7], transforming to a crystalline structure on heating. According to the known empirical rules, a high glass-forming ability can be achieved in complex materials composed from a number of dissimilar elements [8]. Despite the fact that the compositional requirements for martensitic transformations and easy amorphization are different and contradictory, the pseudobinary alloys developed by the substitution of Zr or Hf for Ti and Cu for Ni are good candidates to exhibit both phenomena. Thus, they can serve as amorphous precursors for crystalline shape memory materials with special properties.

Ternary TiNiCu, ZrNiCu, HfNiCu [9,10] and quaternary TiZrNiCu [11,12] alloys were found to show relatively low critical cooling rates required to avoid the formation of crystals and rather wide super-cooled liquid regions between the glass transition and crystallization temperatures. Amorphous–crystalline ribbons of $\text{Ti}_{50}\text{Ni}_{25}\text{Cu}_{25}$

* Corresponding author. Tel.: +380 44 424 14 20; fax: +380 44 424 25 61.
E-mail address: pasko@imp.kiev.ua (A. Pasko).

composition produced by melt-spinning and interesting due to their shape memory properties, were the subject of numerous investigations of the crystallization processes, the resulting microstructure and martensitic transformations [13–17]. Melt-spun TiNi-based ribbons with additions of Cu, Zr, Hf and other elements were also studied [18–21]. The present paper describes research in the thermophysical properties of some rapidly solidified multicomponent (Ti,Zr,Hf,Nb)_A(Ni,Cu,Co,Pd,Ag,Al)_B alloys, as well as the effect of a crystallization thermal treatment on the characteristics of martensitic transformations.

2. Experimental

Four series of TiNi-based alloys that were selected for studying are presented in Table 1. Series A and B are quaternary (TiHf)_A(NiCu)_B alloys with A:B ratio equal to 50:50 and 55:45, respectively, and a wide range of Cu–Ni substitution. The chosen concentration of Hf is known to raise the martensitic transformation temperatures in the ternary system. Series C is based on the proven Ti₅₀Ni₂₅-Cu₂₅ composition, series D (except D7) contains its 55:45 modifications. The multicomponent series C and D are alloyed as follows: Zr, Hf, and Nb substituted for Ti; Co and Pd substituted for Ni; Ag and Al substituted for Cu.

Master alloys have been produced by induction melting in a water-cooled copper crucible under a helium atmosphere (in ‘levitation’) prior to casting in a water-cooled copper mold. Rapidly solidified ribbons were obtained using an advanced variant of melt-spinning technique

known as the planar flow casting. In this method, the nozzle has a narrow slot (0.3–0.35 mm wide and 7–10 mm long), and a small gap between the wheel and nozzle is maintained (0.15–0.2 mm) so that the upper surface of the melt puddle is constrained. Such geometry facilitates a higher stability of the melt flow and allows production of wider ribbons of good quality. Ingots of 5–10 g are melted in the quartz nozzle under helium and ejected on a copper-based wheel rotating at 19 m/s. According to the melt viscosity required for easy ejection, an overheating of 100–150 K above the liquidus temperature is applied; the quenching temperature T_q is controlled by a pyrometer as indicated in Table 1. Thermophysical properties and thermal stability, phase transformations in amorphous and crystalline phases have been studied by using Mettler-Toledo DSC822e differential scanning calorimeter (DSC). Crystal structures were examined at room temperature by X-ray diffraction (XRD) with Co K α radiation. For microstructure observations a Jeol transmission electron microscope (TEM) operating at 120 kV was used. Thin foils were prepared by double-jet electropolishing in a cooled solution of nitric acid in methanol.

3. Results

Most rapidly solidified ribbons are fully amorphous (Table 1), XRD spectra have wide peaks specific for amorphous phase. Exceptions are ribbons C with different proportions of crystalline and amorphous components, and ribbon A1 undergoing a martensitic transformation near

Table 1
Chemical composition of alloys, quenching temperature T_q , and atomic structure (amorphous am, crystalline cr) of rapidly solidified ribbons

Alloy	Nominal composition (at.%)										A:B	T_q (K)	XRD
	Ti	Zr	Hf	Nb	Ni	Cu	Co	Pd	Ag	Al			
A1	32	–	18	–	45	5	–	–	–	–	50:50	1570	am + cr
A2	32	–	18	–	35	15	–	–	–	–	50:50	1560	am
A3	32	–	18	–	25	25	–	–	–	–	50:50	1570	am
A4	32	–	18	–	15	35	–	–	–	–	50:50	1470	am
A5	32	–	18	–	5	45	–	–	–	–	50:50	1370	am
B1	40	–	15	–	18	27	–	–	–	–	55:45	1580	am
B2	40	–	15	–	13	32	–	–	–	–	55:45	1570	am
B3	40	–	15	–	8	37	–	–	–	–	55:45	1370	am
B4	40	–	15	–	3	42	–	–	–	–	55:45	1370	am
B5	40	–	15	–	–	45	–	–	–	–	55:45	1370	am
C1	48	2	–	–	23	23	–	2	–	2	50:50	1550	am + cr
C2	47	3	–	–	23	20	–	2	–	5	50:50	1570	am + cr
C3	45	5	–	–	23	20	–	2	1	4	50:50	1590	am + cr
C4	47	–	3	–	23	24.5	–	2	0.5	–	50:50	1590	am + cr
C5	47	–	–	3	23	22	–	2	–	3	50:50	1590	am + cr
C6	48	2	–	–	20	25	5	–	–	–	50:50	1580	am + cr
C7	40	10	–	–	20	25	5	–	–	–	50:50	1570	am
D1	47	–	8	–	18	24.5	–	2	0.5	–	55:45	1570	am
D2	48	7	–	–	18	23	–	2	2	–	55:45	1590	am
D3	48	7	–	–	18	25	2	–	–	–	55:45	1550	am
D4	48	7	–	–	18	22	5	–	–	–	55:45	1590	am
D5	40	15	–	–	18	25.7	–	1	0.3	–	55:45	1570	am
D6	40	–	15	–	18	25.7	–	1	0.3	–	55:45	1570	am
D7	40	–	15	–	11	32	2	–	–	–	55:45	1390	am

room temperature. The glass-forming ability improves from 50:50 to 55:45 compositions. Alloying with Cu favors amorphization, but decreases the martensitic transformation temperatures (series A).

Thermal analysis has been performed by heating up to 973 K, typical DSC curves are presented in Fig. 1. For all ribbons, a classical behavior characteristic of amorphous material is observed: a broad region of exothermic

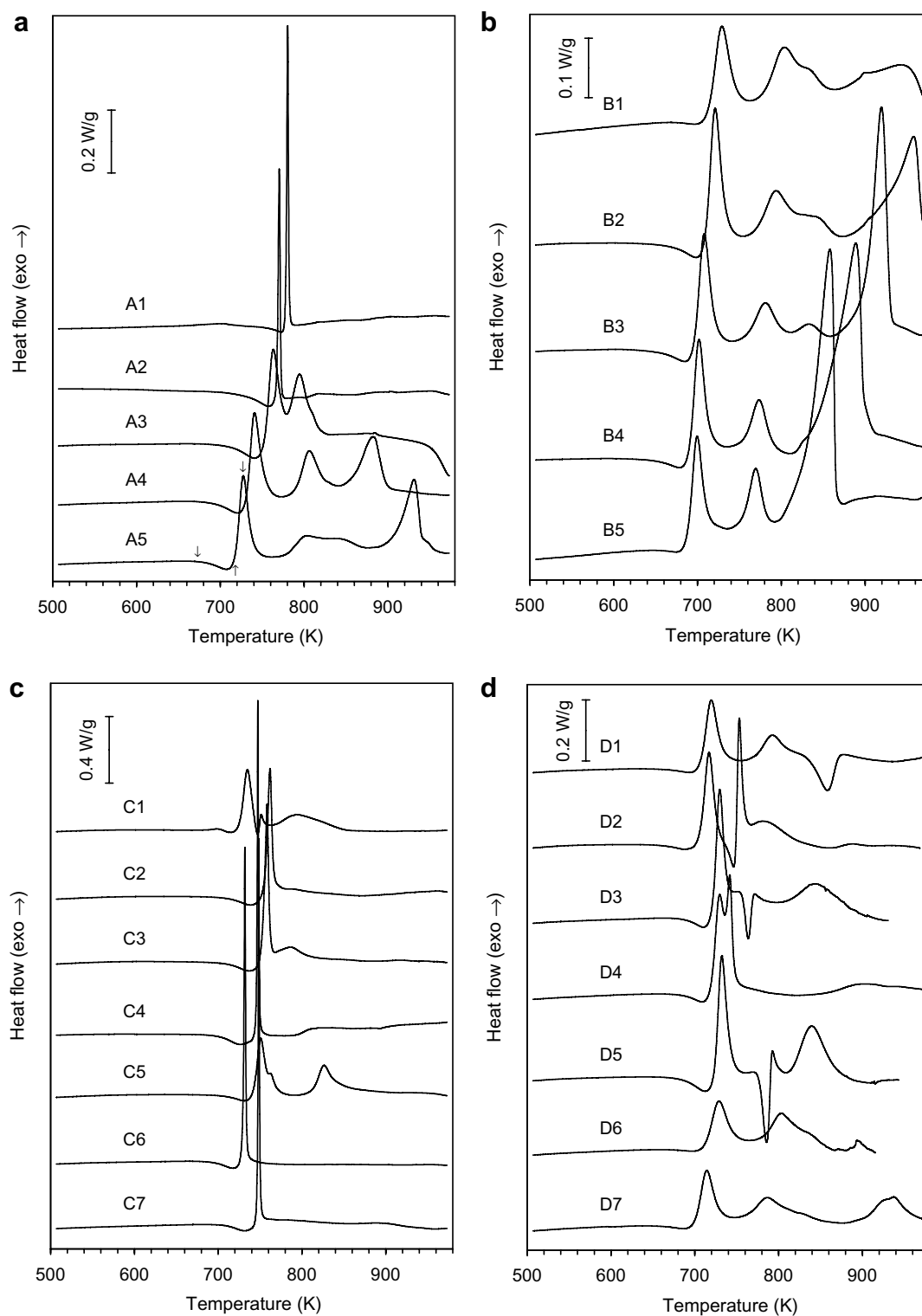


Fig. 1. DSC curves of rapidly solidified ribbons (heating rate 0.17 K/s): the glass transition and crystallization in the amorphous phase. The alloy labels are indicated with temperatures T_g , T_x , T_p for ribbon A5 as an example.

structural relaxation, an endothermic glass transition, and one or several exothermic peaks corresponding to successive transformations from a metastable super-cooled liquid state to the equilibrium crystalline (or quasicrystalline) intermetallic phases at different temperatures. Table 2 contains the glass transition and crystallization temperatures measured at 0.17 K/s heating rate. Temperature T_g is defined here as the calorimetric onset of glass transition, T_x and T_p are the onset and peak temperatures of the first crystallization event. The width of super-cooled liquid region $\Delta T = T_x - T_g$ is believed to be an important characteristic of amorphous material: the greater ΔT the better its glass-forming ability.

Ribbons A with small amount of Cu exhibit one sharp crystallization peak. Addition of more copper leads to broadening and splitting of the heat release event and decrease of the glass transition and crystallization temperatures (Fig. 1(a)). Dependence of ΔT on Cu concentration is not so obvious because of difficulties in correct determination of T_g .

According to XRD data, the investigated A1 ribbon is partially amorphous, and contains also the austenitic B2 and martensitic B19 crystalline phases at room temperature. The TEM observations confirm the presence of an amorphous material in the as-cast state. The crystalline grains form big circular aggregates in the amorphous matrix with a smooth external crystalline–amorphous boundary and wavy internal boundaries between them. Plate-like martensite crystals can be observed inside.

Table 2

Glass transition and crystallization temperatures measured by DSC (heating rate 0.17 K/s)

Alloy	T_g (K)	T_x (K)	T_p (K)	ΔT (K)
A1	738	778	780	40
A2	722	766	769	45
A3	705	752	763	47
A4	684	731	741	47
A5	673	718	727	44
B1	648	712	728	64
B2	656	709	721	53
B3	641	695	707	55
B4	637	691	702	53
B5	640	687	699	47
C1	702	726	736	24
C2	710	756	763	46
C3	708	753	760	45
C4	706	747	752	41
C5	704	742	751	38
C6	697	730	735	33
C7	709	748	753	39
D1	661	707	720	46
D2	660	706	718	46
D3	690	723	731	33
D4	684	719	730	35
D5	684	723	733	39
D6	671	714	730	43
D7	658	701	715	43

Estimated errors: ± 10 K for T_g and ΔT , ± 5 K for T_x , ± 2 K for T_p .

Fig. 2 represents XRD spectra of the fully amorphous ribbon A2 in the as-cast state and after thermal treatment (in a salt bath under helium) at different temperatures. Crystallization leads to appearance of a single cubic phase, which is in agreement with DSC showing a single thermal event of crystallization. The crystallization temperature, determined by XRD for 600 s isothermal annealing (between 743 K and 748 K) is lower than that measured during DSC scan at 0.17 K/s heating rate (766 K).

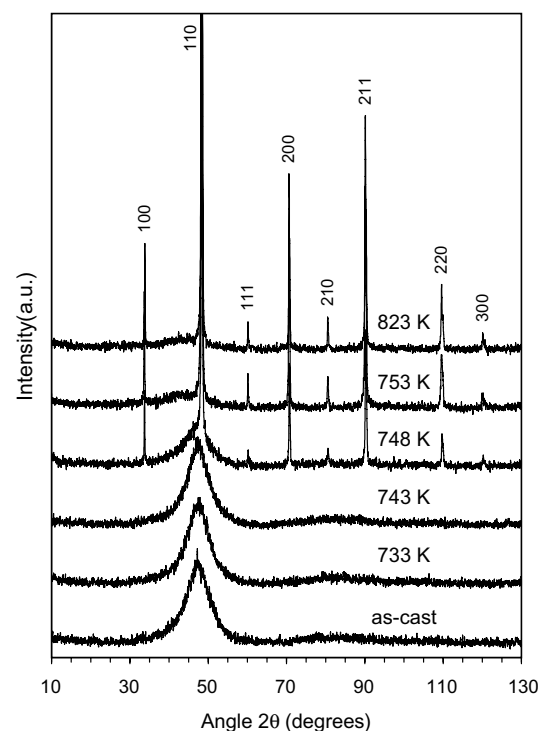


Fig. 2. XRD spectra of ribbon A2 in the as-cast state and after thermal treatment for 600 s at different temperatures: crystallization of the amorphous phase.

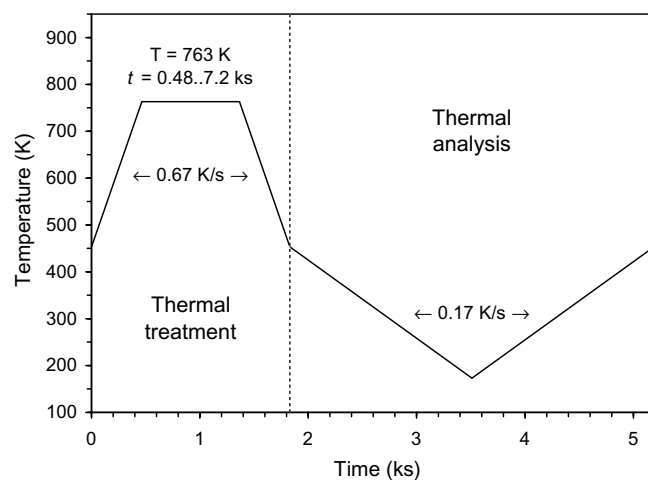


Fig. 3. Calorimeter setup for studying the martensitic transformations in crystallized ribbons, consisting of a thermal treatment (crystallization) cycle followed by a thermal analysis (martensitic transformation) cycle.

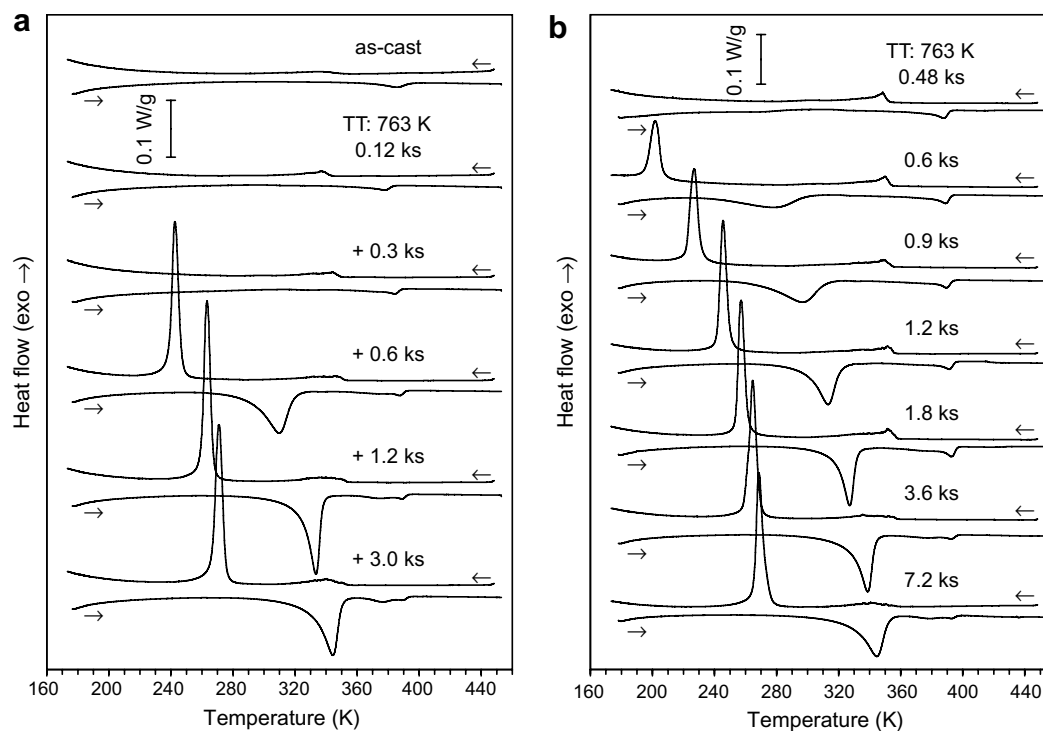


Fig. 4. Martensitic transformation thermal effects in ribbon Al1 after crystallization treatment in calorimeter: (a) one specimen, multiple cycles, cumulative annealing and (b) one cycle, multiple specimens, independent annealing.

It was noticed that after reaching the crystallization point the characteristics of martensitic transformation substantially change. To study the effect of amorphous phase crystallization on the martensitic transformation parameters, a set of experiments on short-term annealing in appropriate temperature range has been carried out. The samples were thermally cycled in the calorimeter, and each run (Fig. 3) was composed from the cycle of annealing (fast heating, staying at a specific temperature and cooling down), and the subsequent cycle of martensitic transformation which was recorded.

The resulting sets of martensitic transformation curves for some important annealing conditions are shown in Fig. 4. The experiments were performed on a single specimen cycled with cumulative thermal treatment (Fig. 4(a)), as well as on several specimens using different independent (without cycling) thermal treatments (Fig. 4(b)). The as-cast ribbon has a comparatively small and wide peak corresponding to the martensitic transformation in the crystalline component of the material. The first stage of annealing leads to sharpening of the original transformation peak. Further annealing causes crystallization of the amorphous component, which reveals itself as appearance and development of a new big peak of martensitic transformation. On continuing thermal treatment, the peak grows and shifts to higher temperatures; subsequently some degradation is observed. The old and new martensitic peaks remain well separated in a wide range of annealing time. The dependence of the peak temperatures of the forward and reverse martensitic transforma-

tions on the time of annealing in calorimeter is presented in Fig. 5.

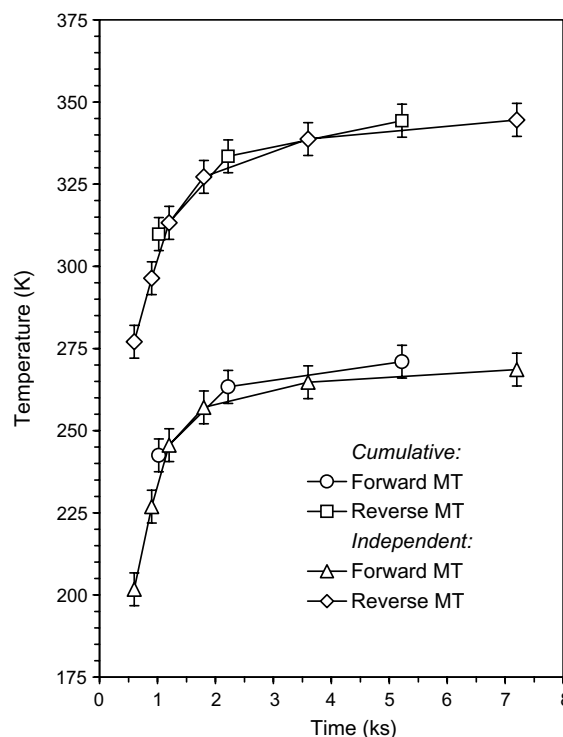


Fig. 5. Dependence of the peak temperatures of the forward and reverse martensitic transformations on the time of cumulative and independent annealing in calorimeter (ribbon Al1).

4. Discussion

Typical microstructure of the annealed A1 ribbon is shown in Fig. 6. The average size of randomly dispersed particles of B2 phase is 100–200 nm, whereas some primary B2 grains formed during rapid solidification are around several microns. The effect of grain size on the martensitic transformation temperatures is a known phenomenon, and even nanoparticles can undergo displacive phase transitions, including thermoelastic martensitic transformations [22].

The appearance of two peaks in ribbon A1 can be explained in such a way that the martensitic transformation

occurs at different temperatures in old big grains and new small grains. The growth of small grains in the amorphous matrix will shift the transformation to higher temperatures, which is observed in the experiment. The process of nucleation of new crystalline grains seems to be fast and difficult for recording. However, some data testify that the new peak of martensitic transformation corresponding to crystallization appears at lower temperatures than the normal one. The TEM observations confirm the presence of small spherical crystalline grains in the amorphous matrix for this stage of annealing. The general tendency of shifting the martensitic transformation to higher temperatures can

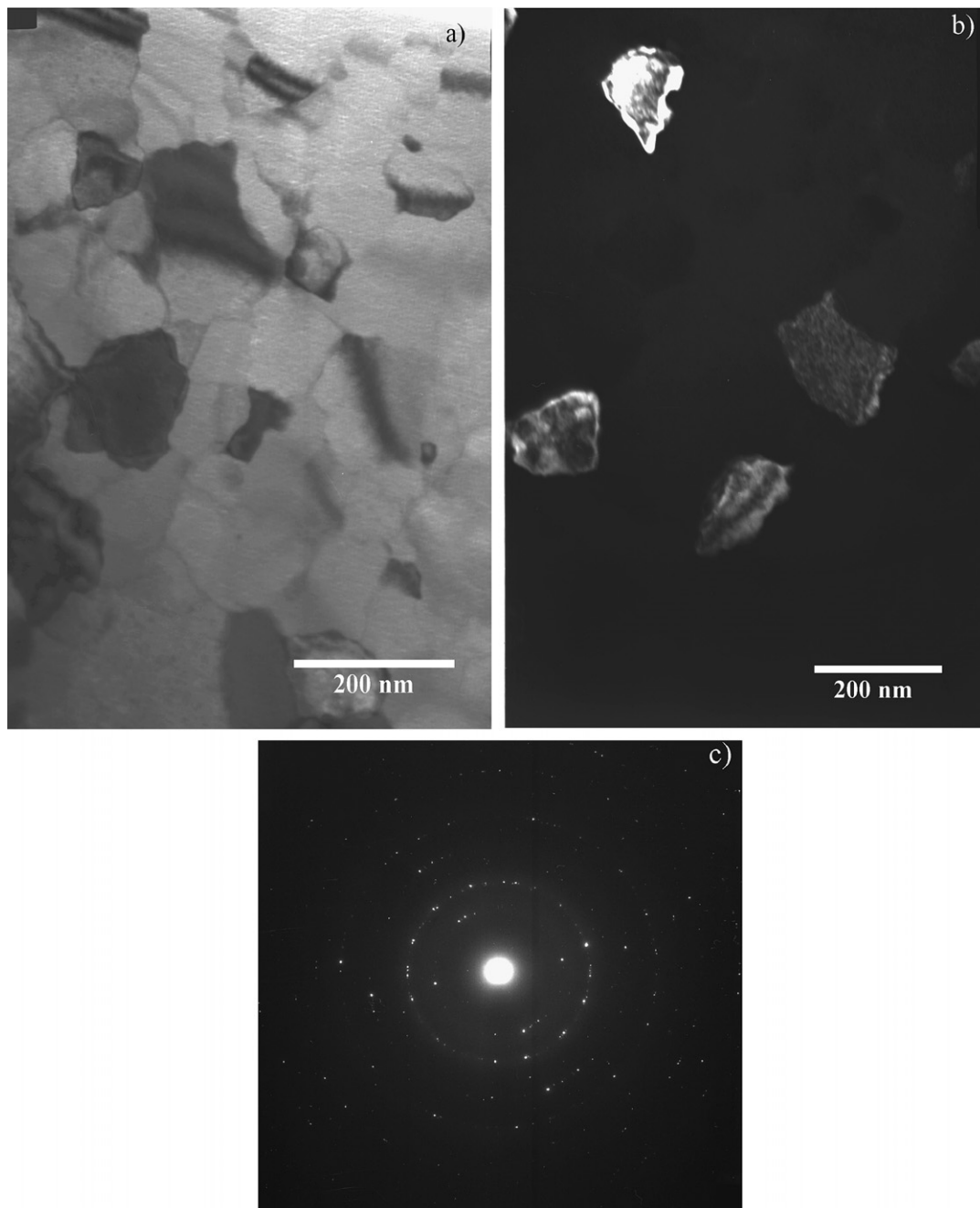


Fig. 6. TEM images of ribbon A1 after annealing at 763 K for 600 s: (a) bright field; (b) dark field; and (c) diffraction pattern.

be attributed to the changes of chemical composition in the transforming matrix.

Martensitic transformations in amorphous–crystalline melt-spun ribbons are affected by a number of factors and can be easily modified applying a special thermal treatment. Depending on the temperature and time conditions, the amorphous component of the material can crystallize in two ways: by nucleation and growth of new small grains or by growth of existing big grains (with possible formation of peculiar internal structure). The precipitates initiate additional inhomogeneity and change the chemical composition. Specific grain size distribution and defect structure control the conditions of martensite crystals formation and in this way determine the temperature profile of martensitic transformations.

5. Conclusion

Partially or fully amorphous ribbons from multicomponent TiNi-based alloys with substitutions of Zr, Hf, Nb for Ti and Cu, Co, Pd, Ag, Al for Ni have been produced by planar flow casting. They crystallize on heating in a single step or show multiple thermal events depending on the chemical composition. The glass transition and crystallization temperatures have been determined by DSC and compared with XRD and TEM data. After appropriate thermal treatment the crystallized material can undergo a martensitic transformation, the characteristic temperatures of which depend on the developed microstructure. Annealing in the temperature interval between the glass transition and beginning of crystallization (as determined by DSC) allows to study the martensitic transformation in new small grains appeared at early stages of crystallization and to compare it with the transformation behavior in the old grains formed during rapid solidification.

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